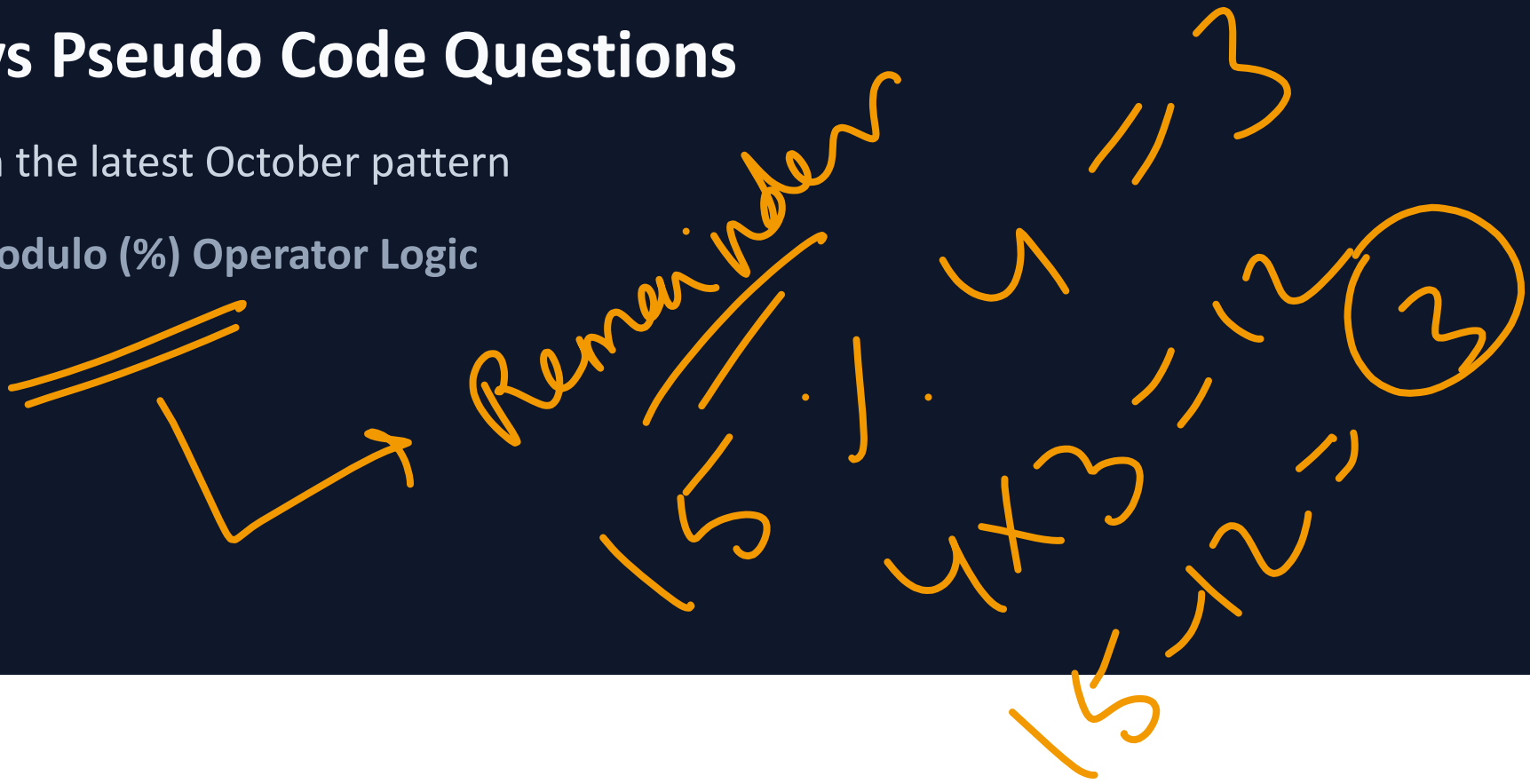
Infosys Pseudo Code Questions

Based on the latest October pattern

Topic: Modulo (%) Operator Logic



Slide 2: Question 1

What will be the output of the following pseudocode?

Plaintext

```
Set Integer num = 456
 Set Integer sum = 0
 while (num > 0) do
   sum = sum + (num % 10)
   num = num / 10
 end-while
 display sum
```

Options:

- **A) 15**
- **B) 12**
- **C) 10**
- **D) 456**



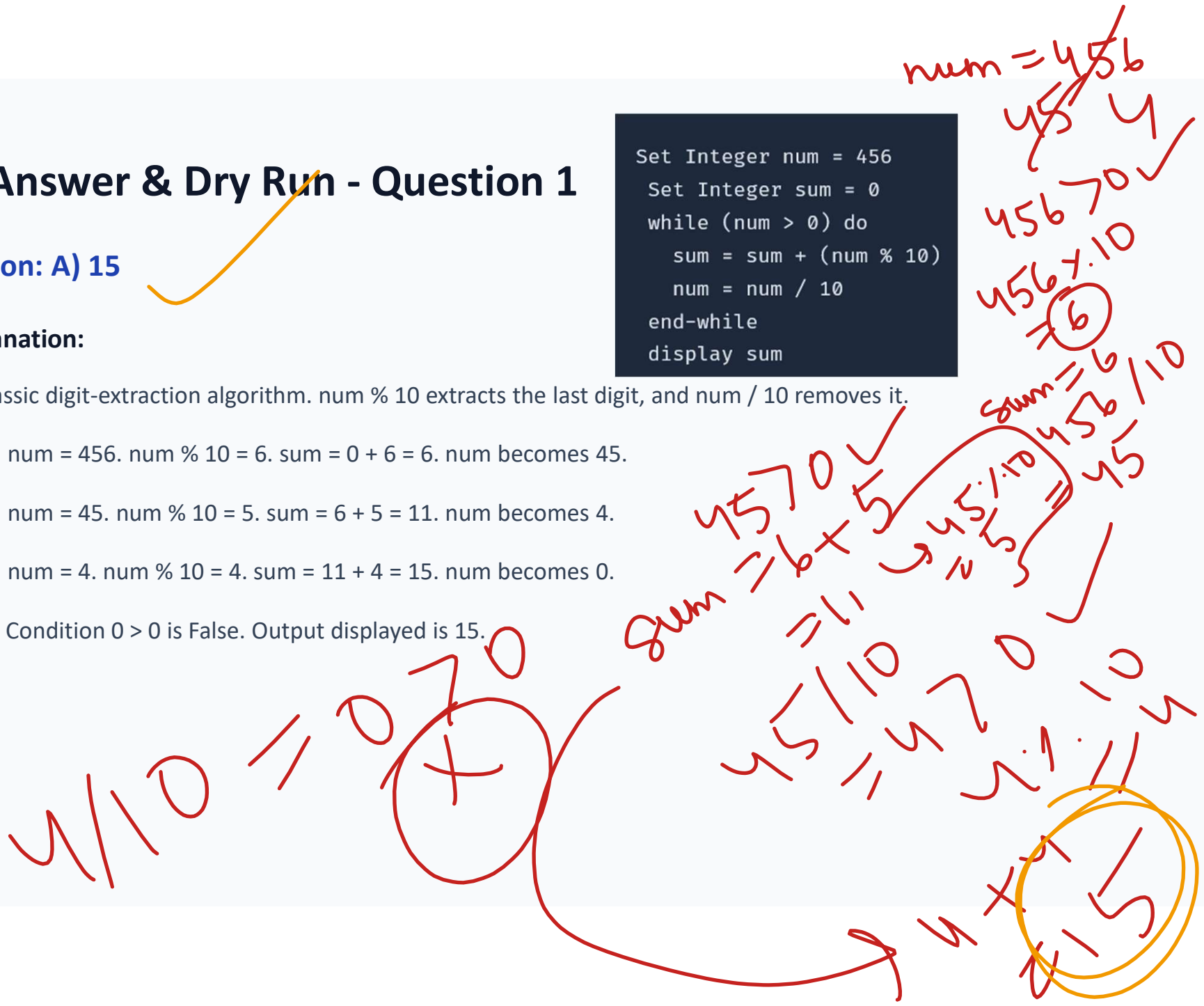
Slide 3: Answer & Dry Run - Question 1

```
Set Integer num = 456
 Set Integer sum = 0
 while (num > 0) do
   sum = sum + (num % 10)
   num = num / 10
 end-while
 display sum
```

Correct Option: A) 15

Dry Run/Explanation:

- This is a classic digit-extraction algorithm. num % 10 extracts the last digit, and num / 10 removes it.
- Iteration 1: num = 456. num % 10 = 6. sum = 0 + 6 = 6. num becomes 45.
- Iteration 2: num = 45. num % 10 = 5. sum = 6 + 5 = 11. num becomes 4.
- Iteration 3: num = 4. num % 10 = 4. sum = 11 + 4 = 15. num becomes 0.
- Loop Ends: Condition 0 > 0 is False. Output displayed is 15.



Slide 4: Question 2

What will be the output of the following pseudocode?

Plaintext

```
Set Integer res = 0
 for i = 1 to 5 step 1 do
   if (i % 2 != 0) then do
     res = res + i
   else do
     res = res - i
   end-if
 end-for
 display res
```

Options:

- **A) -3**
- **B) 3**
- **C) 15**
- **D) 5**

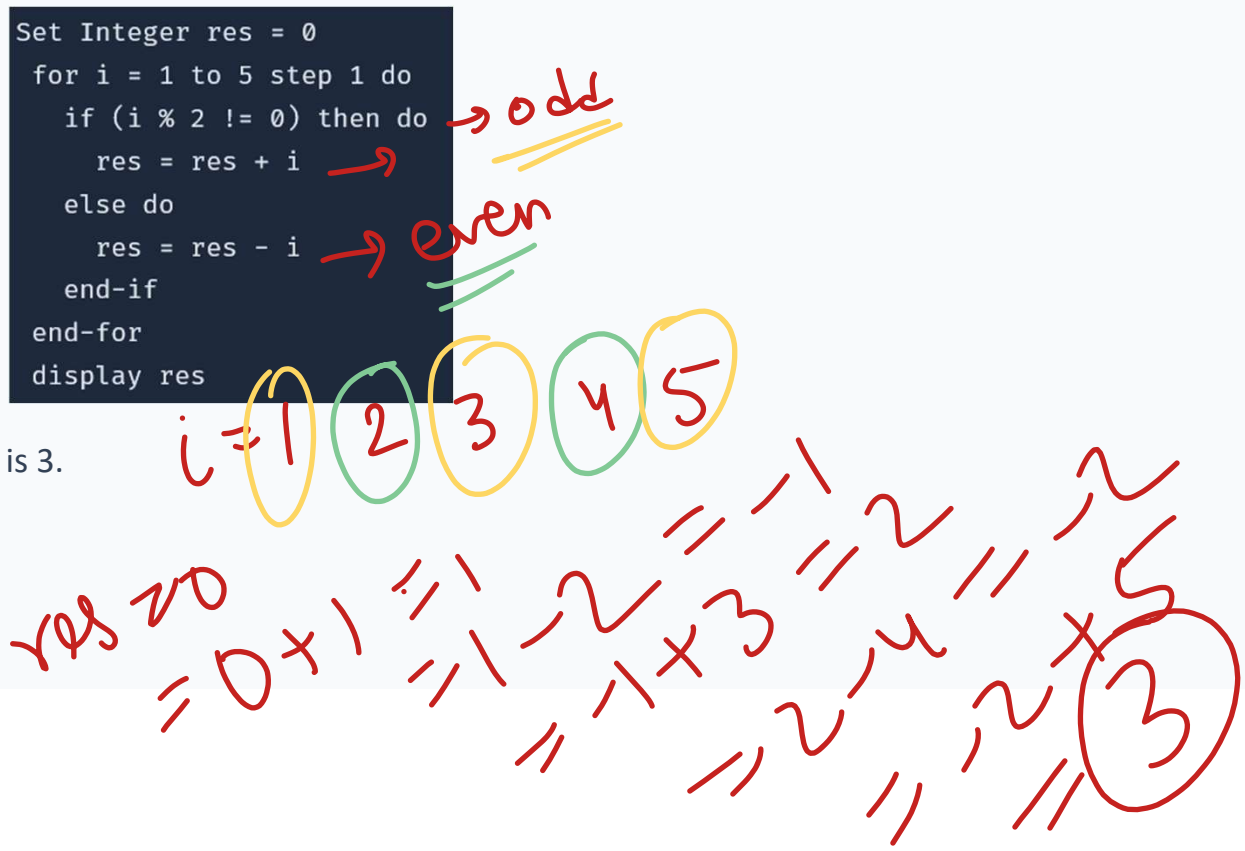
Slide 5: Answer & Dry Run - Question 2

Correct Option: B) 3

Dry Run/Explanation:

- Condition i % 2 != 0 checks if a number is odd. Odd numbers are added, even numbers are subtracted.
- i = 1 (Odd): res = 0 + 1 = 1
- i = 2 (Even): res = 1 - 2 = -1
- i = 3 (Odd): res = -1 + 3 = 2
- i = 4 (Even): res = 2 - 4 = -2
- i = 5 (Odd): res = -2 + 5 = 3
- Loop Ends: The output displayed is 3.

```
Set Integer res = 0
 for i = 1 to 5 step 1 do
   if (i % 2 != 0) then do
     res = res + i
   else do
     res = res - i
   end-if
 end-for
 display res
```



Slide 6: Question 3

What will be the output of the following pseudocode?

Plaintext

```
Set Integer a = 17, b = 5, c = 3
 Set Integer res = (a % b) + (a % c)
 display res
```

Options:

- **A) 2**
- **B) 8**
- **C) 6**
- **D) 4**

Slide 7: Answer & Dry Run - Question 3

Correct Option: D) 4

```
Set Integer a = 17, b = 5, c = 3
Set Integer res = (a % b) + (a % c)
display res
```



Dry Run/Explanation:

- First, evaluate (a % b): 17 divided by 5 leaves a remainder of 2. So, 17 % 5 = 2.
- Next, evaluate (a % c): 17 divided by 3 leaves a remainder of 2. So, 17 % 3 = 2.
- Add the two remainders: res = 2 + 2 = 4.
- The output displayed is 4.

Slide 8: Question 4

What will be the output of the following pseudocode?

Plaintext

```
Set Integer count = 0
 for i = 1 to 10 step 1 do
   if (i % 3 == 0) then do
     continue
   end-if
   count = count + 1
 end-for
 display count
```

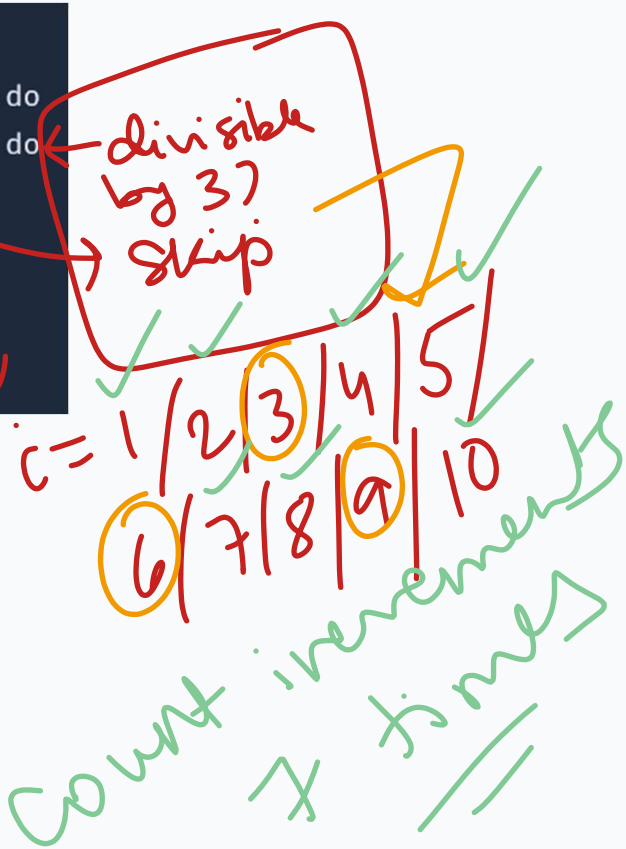
Options:

- **A) 10**
- **B) 7**
- **C) 3**
- **D) 0**

Slide 9: Answer & Dry Run - Question 4

Correct Option: B) 7

```
Set Integer count = 0
 for i = 1 to 10 step 1 do
   if (i % 3 == 0) then do
     continue
   end-if
   count = count + 1
 end-for
 display count
```



Dry Run/Explanation:

- The loop runs from 1 to 10 (total 10 iterations).
- Condition i % 3 == 0 checks if i is a multiple of 3.
- Multiples of 3 between 1 and 10 are: 3, 6, and 9.
- When i is 3, 6, or 9, the continue statement skips the count = count + 1 execution.
- For the remaining 7 numbers (1, 2, 4, 5, 7, 8, 10), count increments normally.
- The output displayed is 7.

Slide 10: Question 5

What will be the output of the following pseudocode?

Plaintext

```
Set Integer x = 23
 Set Integer y = 4
 x = x + (x % y)
 y = y + (x % y)
 display x + " " + y
```

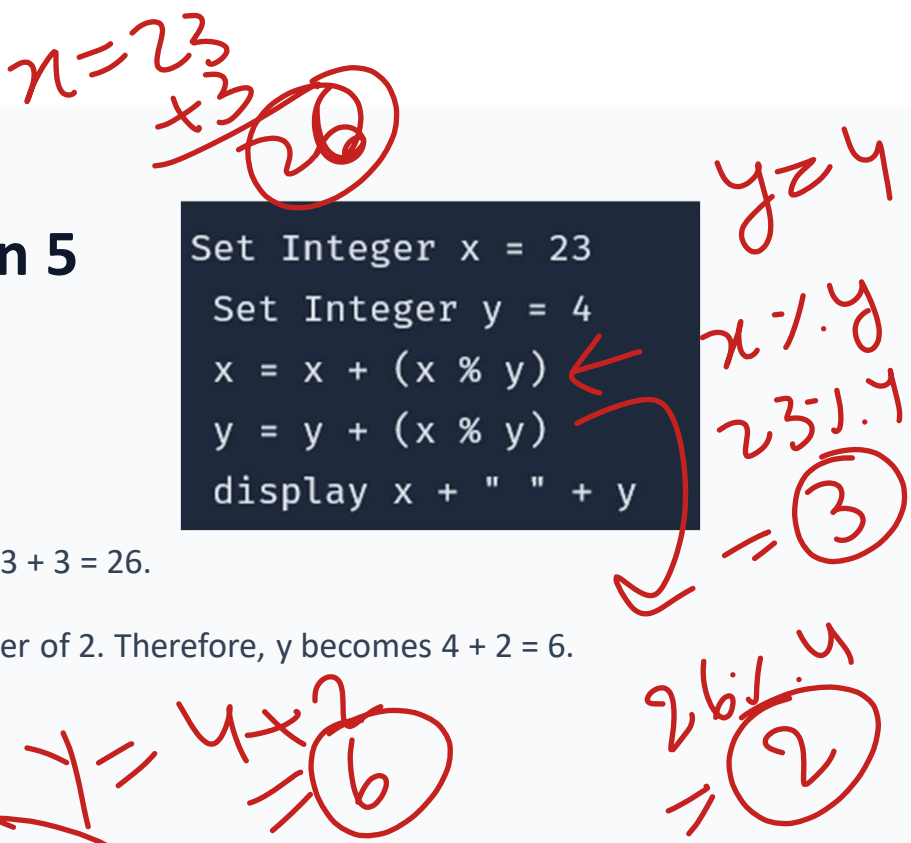
Options:

- **A) 26 4**
- **B) 23 4**
- **C) 26 6**
- **D) 26 5**

Slide 11: Answer & Dry Run - Question 5

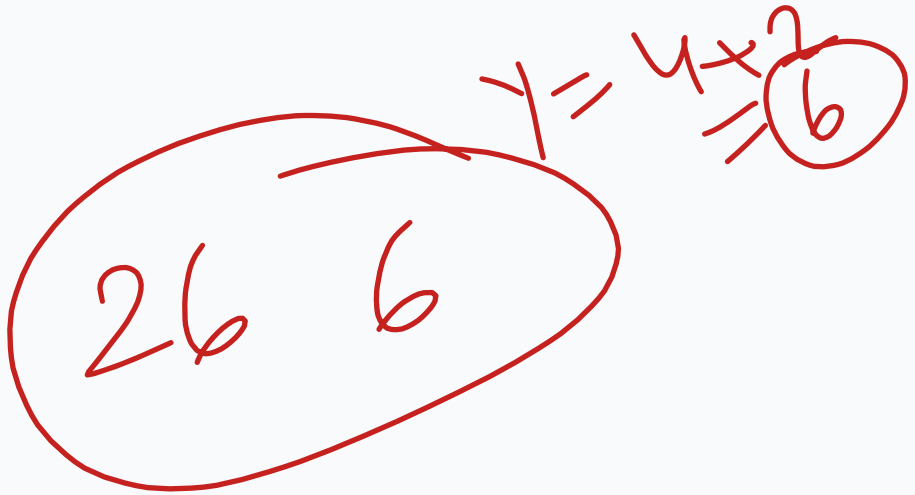
Correct Option: C) 26 6

```
Set Integer x = 23
 Set Integer y = 4
 x = x + (x % y)
 y = y + (x % y)
 display x + " " + y
```



Dry Run/Explanation:

- Line 3: x = x + (x % y). x % y is 23 % 4 = 3. Therefore, x becomes 23 + 3 = 26.
- Line 4: y = y + (x % y). Note: x is now 26. 26 % 4 leaves a remainder of 2. Therefore, y becomes 4 + 2 = 6.
- The output displayed is 26 6.



Slide 12: Question 6

What will be the output of the following pseudocode?

Plaintext

```
Set Integer arr[5] = {10, 20, 30,
40, 50}
 Set Integer res = 0
 for i = 0 to 4 step 1 do
   if (arr[i] % 4 == 0) then do
     res = res + arr[i]
   end-if
 end-for
 display res
```

Options:

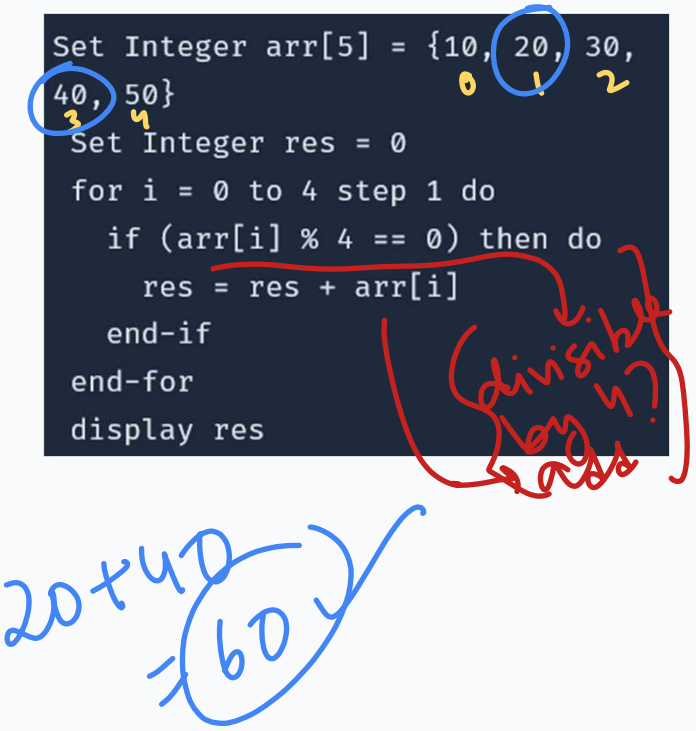
- **A) 150**
- **B) 60**
- **C) 80**
- **D) 40**

Slide 13: Answer & Dry Run - Question 6

Correct Option: B) 60

Dry Run/Explanation:

- Iterates array and adds elements to res only if divisible by 4 (remainder is 0).
- i = 0: 10 % 4 = 2 (False)
- i = 1: 20 % 4 = 0 (True) → res = 0 + 20 = 20
- i = 2: 30 % 4 = 2 (False)
- i = 3: 40 % 4 = 0 (True) → res = 20 + 40 = 60
- i = 4: 50 % 4 = 2 (False)
- The output displayed is 60.



Slide 14: Question 7

What will be the output of the following pseudocode?

Plaintext

```
Set Integer m = 8
 Set Integer n = 3
 Set Integer val = (m++ % n) + (m %
n)
 display val
```

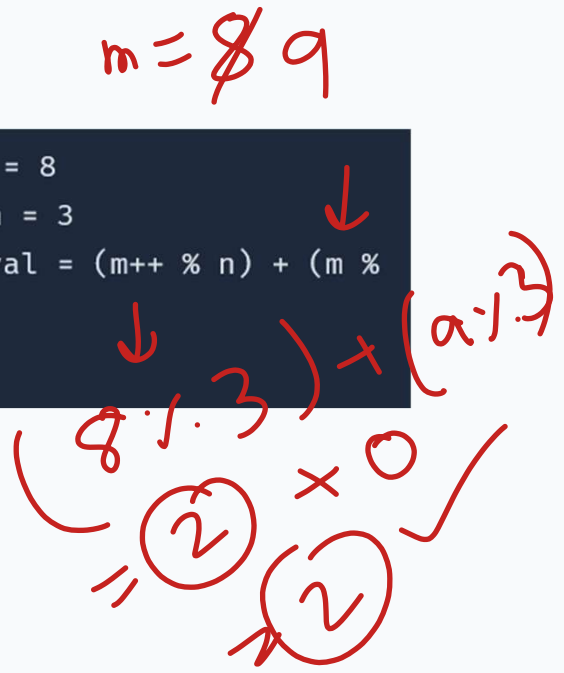
Options:

- **A) 5**
- **B) 4**
- **C) 3**
- **D) 2**

Slide 15: Answer & Dry Run - Question 7

Correct Option: D) 2

```
Set Integer m = 8
 Set Integer n = 3
 Set Integer val = (m++ % n) + (m %
n)
 display val
```



Dry Run/Explanation:

- Evaluates left-to-right. m++ is a post-increment operator.
- (m++ % n) uses current value (8). 8 % 3 = 2. Immediately after, m becomes 9.
- (m % n) uses updated value (9). 9 % 3 = 0.
- val = 2 + 0 = 2.
- The output displayed is 2.

Slide 16: Question 8

What will be the output of the following pseudocode?

Plaintext

```
Set Integer n = 123
 Set Integer rev = 0
 while (n > 0) do
   rev = (rev * 10) + (n % 10)
   n = n / 10
 end-while
 display rev
```

Options:

- **A) 6**
- **B) 123**
- **C) 321**
- **D) 0**

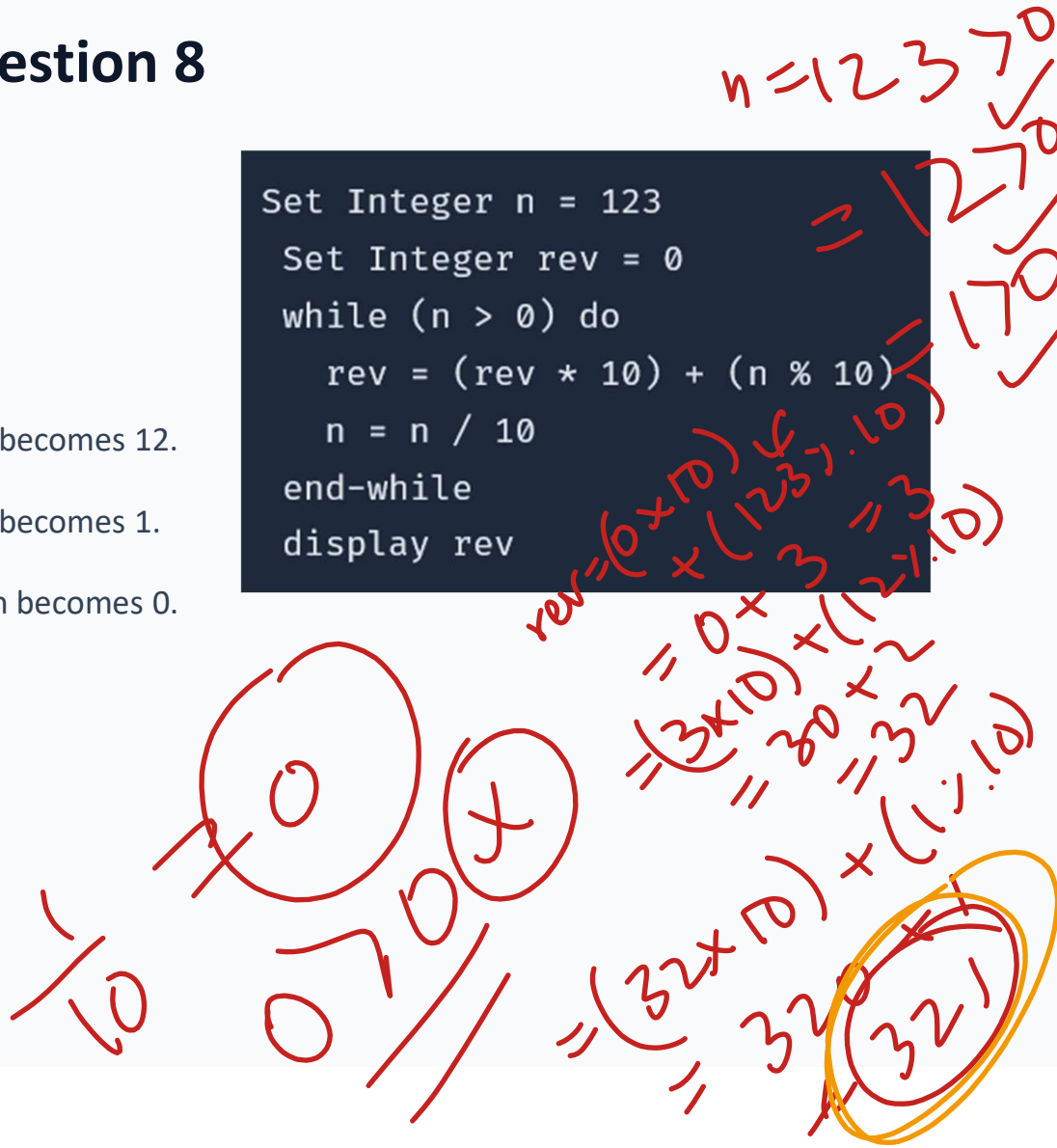
Slide 17: Answer & Dry Run - Question 8

Correct Option: C) 321

Dry Run/Explanation:

- This is the standard algorithm for reversing an integer.
- Iteration 1: n = 123. n % 10 = 3. rev = (0 * 10) + 3 = 3. n becomes 12.
- Iteration 2: n = 12. n % 10 = 2. rev = (3 * 10) + 2 = 32. n becomes 1.
- Iteration 3: n = 1. n % 10 = 1. rev = (32 * 10) + 1 = 321. n becomes 0.
- Loop Ends: Condition 0 > 0 is False. Output is 321.

```
Set Integer n = 123
 Set Integer rev = 0
 while (n > 0) do
   rev = (rev * 10) + (n % 10)
   n = n / 10
 end-while
 display rev
```



Slide 18: Question 9

What will be the output of the following pseudocode?

Plaintext

```
Set Integer num = 12, count = 0
 for i = 1 to num step 1 do
   if (num % i == 0) then do
     count = count + 1
   end-if
 end-for
 display count
```

Options:

- **A) 4**
- **B) 5**
- **C) 6**
- **D) 12**

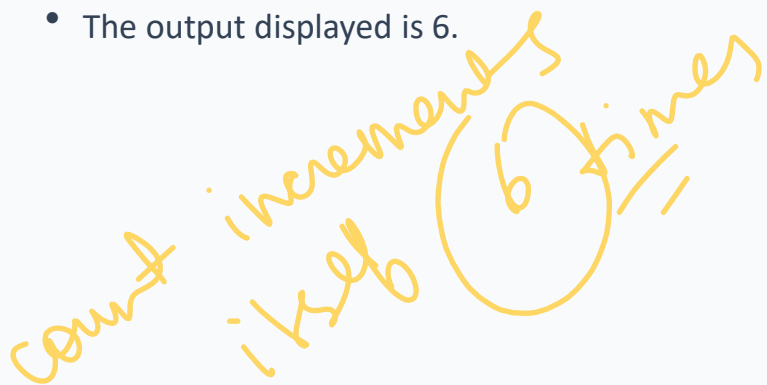
Slide 19: Answer & Dry Run - Question 9

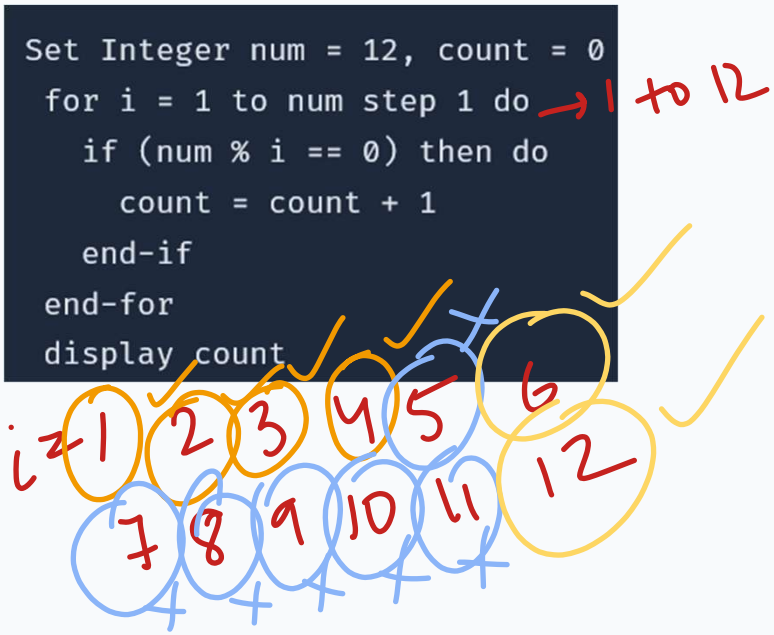
Correct Option: C) 6

Dry Run/Explanation:

- Counts the number of proper factors (divisors) of num. num % i == 0 checks for perfect division.
- The loop runs from 1 to 12.
- Numbers leaving a remainder of 0 when dividing 12: 1, 2, 3, 4, 6, 12.
- The count variable increments exactly 6 times.
- The output displayed is 6.

```
Set Integer num = 12, count = 0
 for i = 1 to num step 1 do
   if (num % i == 0) then do
     count = count + 1
   end-if
 end-for
 display count
```





Slide 20: Question 10

What will be the output of the following pseudocode?

Plaintext

```
Set Integer val = 83
 Set Integer res = (val % 15) % 4
 display res
```

Options:

- **A) 1**
- **B) 2**
- **C) 3**
- **D) 0**

Slide 21: Answer & Dry Run - Question 10

Correct Option: D) 0

```
Set Integer val = 83
Set Integer res = (val % 15) % 4
display res
```

Dry Run/Explanation:

- This tests the nested evaluation of modulo operators.
- First, compute (val % 15). 83 divided by 15 is 5 (15 * 5 = 75), leaving a remainder of 8. Thus, 83 % 15 = 8.
- Next, compute 8 % 4. 8 divided by 4 leaves a remainder of 0.
- res is assigned 0.
- The output displayed is 0.